Aman Gautam

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Experience

DragonX Systems (AIC, IIT Delhi) Jul 2026 – Oct 2026

Founding Software Engineering Intern

- Accelerated OpenROAD timing-repair workloads using CUDA and incremental static timing analysis, achieving 15.7× faster GCD timing repair and 62% lower Ibex repair runtime, reducing compute time for physical-design optimization.
- Automated hardware-specification generation using the OpenAI API, converting natural-language requirements into validated design-canvas representations and reducing manual specification-to-design steps.
- Built a hardware design-space exploration pipeline to compare architectural configurations across performance, power, and efficiency metrics, reducing manual evaluation effort.

Open Source Contributor *Portfolio*

20+ merged PRs across 8 production-grade ecosystems

- Contributed to Apache RocketMQ, Armeria (LINE), Spring Kafka, Gradle, cBioPortal, OpenRewrite, and InsForge.
- **Apache RocketMQ:** Fixed broker generating invalid POP checkpoints for non-existent consumer groups (#9982).
- **Armeria (LINE):** Fixed IDN-normalization failure on long hostnames with ports, and a sparse-map NullPointerException in *Http1ObjectEncoder* (#6860, #6899).
- **cBioPortal:** Fixed REST API returning HTTP 500 instead of 404 for missing reference genome genes via defensive null handling (#11910).

Personal Projects

Real-Time Collaborative Sync Engine | *GitHub*

- Engineered a distributed document synchronization engine using Spring Boot, WebSockets, and Redis Pub/Sub, delivering real-time updates across connected clients and eliminating manual document refresh.
- Implemented Kafka-backed event persistence and automated PostgreSQL snapshots every 100 operations, enabling state recovery through snapshot restoration and incremental event replay instead of full-history reconstruction.
- Built consistent-hash routing across 3 configured application nodes with Operational Transformation (OT) for concurrent edits, supporting distributed request processing and reducing manual conflict-resolution requirements.

NyayaSetu – Privacy-First Legal RAG Search Engine | *Demo Video* | *GitHub*

- Built a legal search pipeline using Apache Tika, Tess4J OCR fallback, 768-dimensional embeddings, and Milvus to retrieve relevant judgment passages.
- Integrated Spring Boot and local Llama 3.1 inference to generate context-aware answers from retrieved legal documents.
- Reduced legal judgment analysis time from 60 to 5 minutes (91.7%) in evaluated workflows.
- *Technology Used: Spring Boot, Apache Tika, Tess4J, Milvus, MySQL, Ollama (Llama 3.1), React, Vite*

CODOVYN — Multi-Agent AI Development IDE | *GitHub*

- Developed a standalone macOS development IDE integrating AI coding agents, persistent task history, model-routing policies, and task recovery workflows to reduce manual agent-management overhead.
- Implemented isolated Git worktree editing, reviewed patch application, durable task checkpoints, and policy-controlled agent handoffs to protect source-code integrity during AI-assisted development.
- *Technology Used: JavaScript, Node.js, SQLite, Git, Code-OSS*

Technical Skills

Languages: Java, C++, SQL
Backend & Frameworks: Spring Boot, REST APIs, Microservices, WebSockets, Hibernate/JPA
Distributed Systems & Messaging: Apache Kafka (Message Queues), Redis (Caching), Multithreading & Concurrency
AI/ML: Retrieval-Augmented Generation (RAG), LLM Integration (OpenAI API, Ollama/Llama 3.1), Vector Embeddings
DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS (EC2), Render, Netlify , Swagger/OpenAPI
Databases: PostgreSQL, MySQL, Milvus (Vector DB)

Education

Galgotias College of Engineering and Technology, Greater Noida 2023 – 2027

Bachelor of Technology in Computer Science and Engineering

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, System Design

Achievements

- Competitive Programming: LeetCode Knight, Max Rating 2102 (Top 2%) (Gautam-aman); Codeforces Specialist, Max Rating 1434 (AmanGautam_28); 2000+ DSA problems solved across platforms.
- Winner, Techfest Hackathon 2026: Built Smart-Sagar, an AI-powered maritime solution, among competing student teams.
- Merit-Based Scholarship: Awarded full tuition fee waiver for top rank in B.Tech admission merit list, Galgotias College (2023).